A High-Radix Mass Storage System

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Title: A High-Radix Mass Storage System

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Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "A Method and System for High-Radix Physical Layer Data Encoding and Representation" and "A Memory System for the Storage and Retrieval of Parametric and Symbolic Data Representations." The present invention focuses on the physical implementation of mass storage devices utilizing high-radix states, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A mass storage system configured to physically store data using a high-radix encoding scheme, where each storage element (e.g., a memory cell, a magnetic domain) can reliably represent one of N discrete, distinguishable physical states, with N being an integer greater than two. This system comprises a high-radix storage medium (e.g., flash memory, magnetic media, optical media) and a high-radix storage controller. The controller is configured to write abstract symbols directly into the storage medium by setting the physical state of corresponding storage elements and to read data by detecting these states and converting them back into abstract symbols. This architecture fundamentally moves beyond binary data

storage, enabling higher data density per physical element and a more direct representation of information for parametric computing systems, reducing the need for binary conversion at the storage interface.

Background of the Invention

Mass storage systems, such including Hard Disk Drives (HDDs), Solid State Drives (SSDs), and optical discs, are the backbone of data persistence. While these technologies have advanced tremendously in density and speed, their fundamental operational principle remains binary. Data is ultimately stored as a series of magnetic orientations, electrical charges, or optical reflections, each representing a binary 0 or 1. Even multi-level cell (MLC) technologies in flash memory, which store more than one bit per cell, do so by defining multiple voltage thresholds to represent combinations of binary digits. This binary foundation limits the intrinsic data density of each physical storage element and necessitates continuous binary conversion when interacting with advanced, non-binary computing paradigms.

Summary of the Invention

The present invention introduces a mass storage system that natively stores data in high-radix physical states, providing increased intrinsic data density per storage element and enabling direct compatibility with high-radix and parametric computing environments. This system represents a fundamental shift in how data is physically encoded and stored.

The system comprises: (1) **A High-Radix Storage Medium:** This is a physical medium (e.g., a specially designed flash memory cell, a novel magnetic material, an advanced optical medium) where each fundamental storage element can reliably hold one of N discrete, distinguishable physical states, with N greater than 2. For instance, a flash cell might reliably hold 16 distinct charge levels, directly representing a radix-16 symbol, rather than merely encoding 4 bits (as in QLC flash). (2) **A High-Radix Storage Controller:** This controller interfaces with the high-radix storage medium. It is configured to directly translate incoming abstract symbols into the corresponding N physical states for writing. Conversely, upon reading, it detects the N physical states from the storage elements and translates them back into abstract symbols.

This architecture bypasses the need for binary-to-radix conversion at the storage interface, streamlining data flow for parametric CPUs and memory systems. It offers significantly

higher data density per physical element, extending the lifespan and capacity of mass storage devices, and provides a native environment for storing perfectly precise parametric data without approximation.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram comparing traditional binary/MLC data storage with high-radix mass storage.

FIG. 2 is a block diagram of a high-radix mass storage system, illustrating the controller and storage medium.

Detailed Description of the Invention

Referring to **FIG. 1**, a conceptual comparison is made between traditional storage (100) and high-radix storage (150). In traditional storage (100), a physical cell (110) stores a binary value (e.g., '0' or '1') or encodes multiple bits using multiple voltage levels (MLC). A high-precision number (115) must be approximated into binary (118) before being stored in such cells. In high-radix storage (150), a physical cell (160) directly stores one of N discrete, distinguishable physical states (165), where N greater than 2. The high-precision number (115) is directly mapped to a sequence of these N-ary states without intermediate binary approximation.

FIG. 2 illustrates a block diagram of a high-radix mass storage system (200). A High-Radix Storage Controller (210) interfaces with a High-Radix Storage Medium (220). The controller receives abstract symbols (205) for writing. It translates these symbols into the

corresponding N physical states and writes them to the storage medium (220) via a high-radix physical interface (215). When reading, the controller detects the physical states from the medium via the interface (215) and translates them back into abstract symbols (225) for output. This system provides a native, non-binary storage solution.

Claims

What is claimed is:

1. A mass storage system, comprising:
    - a high-radix storage medium comprising a plurality of storage elements, wherein each storage element is configured to reliably hold one of N discrete, distinguishable physical states, where N is an integer greater than two; and
    - a high-radix storage controller communicatively coupled to the high-radix storage medium, the controller configured to:
        - receive abstract symbols for writing;
        - translate said abstract symbols into corresponding physical states from the N discrete states; and
        - cause the high-radix storage medium to store said physical states in its storage elements.
2. The mass storage system of claim 1, wherein the high-radix storage controller is further configured to:
    - read physical states from the storage elements of the high-radix storage medium; and
    - translate said physical states back into abstract symbols.
3. The mass storage system of claim 1, wherein the storage elements are flash memory cells configured to store N distinct charge levels.
4. The mass storage system of claim 1, wherein the storage elements are magnetic domains configured to store N distinct magnetic orientations.
5. The mass storage system of claim 1, wherein the abstract symbols represent unevaluated mathematical expressions.

6. A method for storing data in a mass storage system, comprising:

    - providing a high-radix storage medium comprising storage elements, each element capable of reliably holding one of N discrete, distinguishable physical states, where N is an integer greater than two;
    - receiving abstract symbols at a high-radix storage controller;
    - translating said abstract symbols into corresponding physical states from the N discrete states;
    - causing the storage elements of the high-radix storage medium to assume said physical states, thereby storing the data; and
    - subsequently, reading the physical states from the storage elements and translating them back into the abstract symbols.

7. The method of claim 6, wherein the received abstract symbols are components of an unevaluated parametric data structure.

8. The method of claim 6, wherein N is 16 or greater.

Drawings

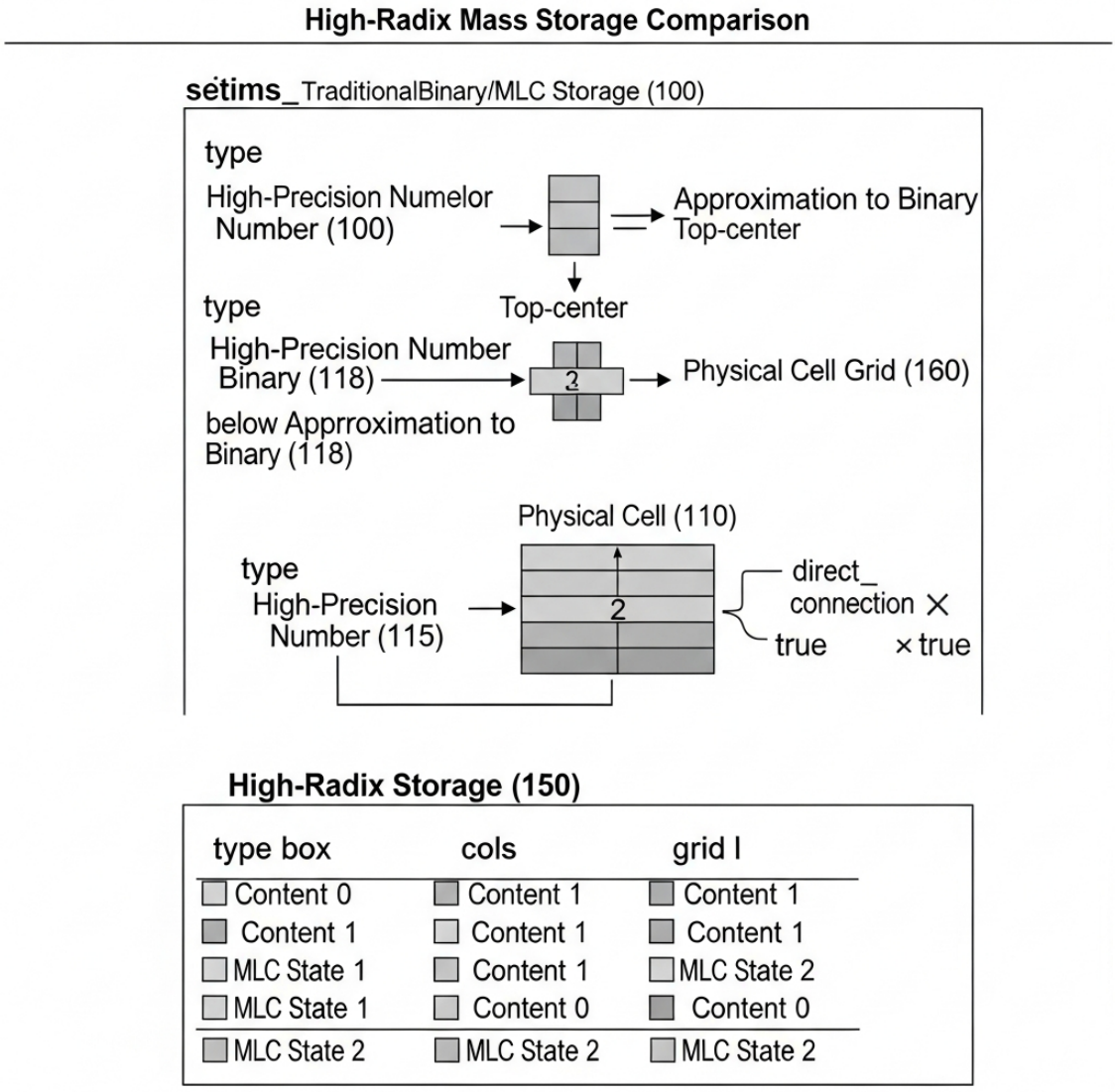


Figure 1: Conceptual diagram comparing traditional binary/MLC data storage with high-radix mass storage.

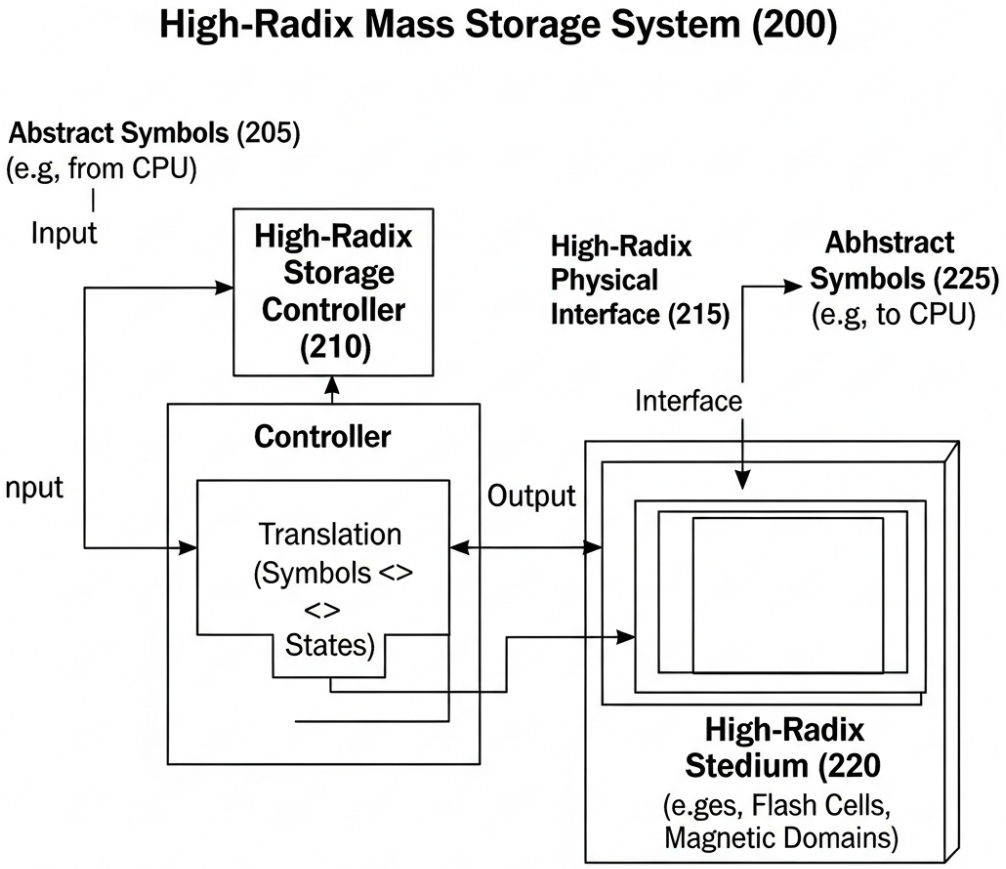


Figure 2: Block diagram of a high-radix mass storage system, illustrating the controller and storage medium.